Example Laboratories

subject: **Balancing the Last Page of a Two-Column Article**

Case 2026-14 — File 2026-14-3

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from: **A. Typesetter**
Room 2C-518 x6021
B. Compositor
Room 2C-521 x6044

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MEMORANDUM FOR FILE

Abstract The last page of an article set in two columns is balanced: its first column ends at half the height of its text, so that both end level. The height is measured on one pass and used on the next.

1 The problem

Troff sets a page's columns one after the other: the first until it is full, then the second. On the last page of an article the text may fill the first column and only a few lines of the second, which looks unfinished, and journals do not print pages so.

2 The method

The height of the text on the last page is not known until it has been set. The macros record it, at the end of the document, in the file of cross-references that the book command already reads on the next pass. On that pass, when the last page begins, the first column's foot is moved to half the recorded height, rounded up to a whole line, and the second column takes the rest.

A page balanced so would measure differently on the next pass, since the first column now ends early. The height is therefore recorded once, from the page as it was set unbalanced, and kept on later passes; the passes then agree, and the command stops.

3 Results

On the example article distributed with the macros, and on a generated one of six pages with notes and figures in both columns, the last page's columns end within a line or two of each other, and no word of the text overlaps another.

document	pages	words overlapping
article.ms	2	0
generated	6	0

A. Typesetter
B. Compositor